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Does the Firm-Level Variance Risk Premium Predict Short-Variance Returns? Evidence from U.S. Large-Cap Equities

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Abstract

This thesis studies whether the firm-level variance risk premium (VRP) predicts the returns of short-variance strategies in U.S. large-cap equities. The analysis uses a hand-picked panel of 73 prominent U.S. firms between 1996 and 2024, including ten firms that failed or were severely distressed. The total VRP positively and significantly predicts the next-month short-variance payoff (0.204, $t = 5.60$), although the R^2 is only 2.2% and the pooled slope is materially influenced by the 2008-2009 crisis, whereas the Fama-MacBeth slope remains significant without it. Contrary to the index-level evidence, the downside leg does not drive this predictability. A long-short strategy selling variance in the highest-VRP quintile, and going long variance in the lowest, earns an annualized alpha of 90.2% ($t = 3.20$) after controlling for the Fama-French five factors, momentum and changes in the VIX, with alpha surviving plausible transaction costs and the exclusion of the 2008-2009 crisis period. Overall, the results point to a firm-level premium that is partly harvestable but concentrated in volatile periods and exposed to severe tail losses.

Table of Contents

Abstract ii
1 Introduction 1
2 Theoretical Framework 3
2.1 The Variance Risk Premium 3
2.1.1 Definition and Conceptual Background 3
2.1.2 Early and Seminal Studies 5
2.1.3 Theoretical Explanations for the VRP 6
2.1.4 Cross-Sectional and Predictive Evidence 7
2.1.5 Measuring the Variance Risk Premium: Challenges and Implications 8
2.2 Short-Variance Strategies 9
2.2.1 Definition and Mechanics 9
2.2.2 Short-Variance Returns: Empirical Evidence 10
2.3 The Relationship between the VRP and Short-Variance Strategy Returns 11
2.3.1 Predictive Power of the VRP 11
2.3.2 Risk-Adjusted Performance and the Fama-French Framework 12
2.3.3 Research Expectations 12
3 Data 13
3.1 Sample 13
3.2 Variables 15
3.2.1 Implied Variance Measures 15
3.2.2 Realized Variance Measures 16
3.2.3 Variance Risk Premia 17
3.2.4 Short-Variance Strategy Payoffs 17
3.3 Control Variables 18
3.4 Descriptive statistics 18
4 Method 20
4.1 Estimation Strategy 20
4.2 Panel Predictive Regressions – H_1 and H_2 20
4.3 Subsample Estimation – H_3 21
4.4 Trading Strategy and Risk-Adjusted Performance – H_4 22
4.5 Out-Of-Sample Predictive Ability 23
4.6 Robustness Checks 24
5 Results 25
5.1 The predictive content of the total variance risk premium (H_1) 25
5.2 The Downside Leg and the Total Premium (H_2) 27
5.3 Stability Across Regimes (H_3) 27

5.4 The Trading Strategy and its Risk-Adjusted Performance (H_4) 28
5.5 OOS Predictive Ability 30
5.6 Robustness Checks 31
6 Discussion 36
7 Conclusion 39
7.1 Limitations and Future Research 40
References 42
Declaration on the use of generative AI 46
Appendices 46

1 Introduction

This thesis studies the relationship between the variance risk premium (VRP) and the returns of a short-variance trading strategy in the U.S. equity market. Volatility is not merely a statistical measure of returns; it is a state variable that investors directly take into account. Volatility rises with the magnitude of returns irrespective of their sign, but in equity markets the response is asymmetric: negative returns raise subsequent volatility considerably more than positive returns of comparable size (Black, 1976; Christie, 1982; Bekaert & Wu, 2000). Volatility spikes therefore cluster with market declines, so an asset that gains value when volatility rises tends to pay out precisely when investors are least willing to absorb further losses. Such an asset is worth more to them than its expected cash flows in isolation would justify, and they will accept a negative expected return in order to hold it. In other words, they pay more than the actuarially fair price for protection against volatility. The counterparty selling that insurance must, in equilibrium, be compensated for bearing the losses if the feared risk were to materialize. That compensation is the variance risk premium, measured as the difference between the option-implied expectation of future variance and the variance that subsequently realizes. Short-variance strategies capture this premium by systematically selling volatility, exposing the seller to the risk that realized variance exceeds implied variance. Institutional investors commonly attain such exposure through covered call (buy-write) strategies, which in certain sample periods have outperformed the S&P 500 with lower return volatility (Hill et al., 2006; Israelov & Nielsen, 2015). Studying this relationship at the individual-stock level is relevant as it provides insight into whether the variance risk premium translates into economically meaningful returns for an implementable, cross-sectional volatility-selling strategy, and whether those returns reflect compensation for bearing systematic risk.

The VRP emerged as an important predictor of asset returns over the past two decades, with Bollerslev et al. (2009) serving as the primary catalyst for this area of research. They demonstrate that the VRP, defined as the difference between model-free implied variance and high-frequency realized variance, contains significant information about future stock returns, especially at the quarterly horizon. The subsequent research that followed is then summarized by Zhou (2018), where two schools of thought emerged. First, the risk-based view, which argues that the VRP is a reward for bearing crash or jump risk (Carr & Wu, 2009; Drechsler & Yaron, 2011). Second, a demand-based perspective attributes the premium to structural order-

flow differences in option markets combined with limits to arbitrage that prevent the spread from being fully competed away (Bollen & Whaley, 2004; Garleanu et al., 2009). To investigate whether these returns are truly direction-agnostic, Bakshi & Kapadia (2003) built delta-hedged option portfolios to isolate volatility risk. They find negative average returns, which is consistent with the volatility risk being priced. Bali & Zhou (2016) further show that the VRP helps explain cross-sectional return differences beyond traditional factors such as size and momentum. However, as Jiang and Tian (2005) highlight, the results in this area can be sensitive to the characteristics of the selected options, particularly with respect to moneyness and maturity.

Despite the broad consensus in the literature that the VRP contains predictive power for aggregate equity returns, it is still unclear whether the single-stock VRP (measured firm-by-firm) translates into economically meaningful returns for a direct, implementable short-variance strategy. The bulk of existing studies treat the VRP as a macroeconomic predictor for market returns or examine cross-sectional sorts on VRP-related measures without constructing the corresponding volatility-selling payoff. This thesis addresses this gap by changing both the unit of analysis and the outcome variable. Instead of testing if VRP predicts the aggregate market return, this study constructs the VRP stock-by-stock and asks whether it predicts the payoff of a short-variance position in that same stock. The payoff is positive when realized variance falls below implied variance, which happens in most months but is far from guaranteed in any individual stock-month. This leads to the central research question of this thesis: *Does the firm-level variance risk premium predict the risk-adjusted returns of short-variance strategies in U.S. large-cap equities?*

To answer the research question, this study employs an empirical panel analysis on a sample of 73 prominent large-cap U.S. equities across sectors, 10 of which failed, were acquired under distress, or were otherwise severely impaired during the sample period. The inclusion of failed firms is deliberate: a short-variance strategy is most vulnerable precisely when substantial negative events occur, so leaving these firms out would omit the observations in which a high VRP is most likely to generate a trading signal. Data is drawn from Wharton Research Data Services (WRDS) over the period 1996-2024, with option-implied variance taken from the OptionMetrics 30-day volatility surface, and realized variance is constructed from the trailing 21 trading days of Center for Research in Security Prices (CRSP) daily returns. The main test is a panel predictive regression of monthly payoffs on the stock-level VRP, complemented by a Fama-MacBeth cross-sectional procedure, quintile sorts on the portfolio are evaluated against the Fama-French five factors, momentum and changes in the CBOE Volatility Index (VIX), an

out-of-sample forecasting exercise, and a set of robustness checks. A key feature of the design should be noted. The predictor and dependent variable are both constructed from option-implied variance of the same firm, so the study identifies whether implied variance is high with respect to the variance that follows. Explanations attributed to the option market therefore remain available, whilst the frictions of the underlying equity market cannot be separately identified, since the short-variance position never holds the underlying equity.

From the theoretical framework, this thesis anticipates a positive relationship between the stock-level VRP and abnormal returns of short-variance strategies. In other words, stocks with higher option-implied variance relative to expected realized variance should generate higher payoffs for variance sellers. Consistent with Bakshi & Kapadia (2003), and Bali & Zhou (2016), who document priced volatility risk at the individual-stock level, this thesis expects the panel beta on VRP to be positive and significant. The more decisive test is whether a long-short strategy that sells variance in the highest-VRP stocks and buys variance in the lowest-VRP stocks generates positive alpha after controlling for the Fama-French five factors, momentum and changes in the VIX. Positive alpha would indicate that the stock-level VRP captures pricing information not accounted for by conventional equity risk factors, thus supporting the market-friction view. Insignificant alpha would suggest that short-variance returns on single stocks are appropriately explained as compensation for systematic risks already embedded in the five-factor model, consistent with the risk compensation view. By linking the statistical predictability of the single-stock VRP directly to the economic profitability of an implementable, cross-sectional short-variance strategy, this thesis contributes to the literature by bridging aggregate return-predictability evidence of Bollerslev et al. (2009) with the cross-sectional, firm-level volatility pricing evidence of Bakshi & Kapadia (2003), and Bali & Zhou (2016).

2 Theoretical Framework

2.1 The Variance Risk Premium

2.1.1 Definition and Conceptual Background

The variance risk premium (VRP) is a concept in asset pricing that captures the compensation investors require for bearing the risk associated with future variance. Formally, the VRP is defined as the difference between the risk-neutral expectation of future return variance, proxied by implied variance, and the actual (statistical) expectation of future realized

variance (Bollerslev, Tauchen, & Zhou, 2009; Carr & Wu, 2009). When implied variance exceeds realized variance, the VRP is positive, indicating that investors are willing to pay a premium for protection against volatility risk. This spread between the two variance measures reflects investors' aversion to uncertainty, and demand for insurance against adverse market movements.

The VRP should not be interpreted as a pure measure of variance risk compensation. In addition to reflecting compensation for bearing volatility risk, it may also capture time-varying risk aversion, macroeconomic uncertainty, and demand for tail-risk hedging in option markets (Carr & Wu, 2009; Drechsler & Yaron, 2011; Zhou, 2018). Because of this composite nature, the VRP fluctuates with broader economic conditions and investor sentiment, not just with the statistical properties of volatility itself. Although much of the early literature focuses on the aggregate VRP derived from index options, the same logic applies at the single-stock level: each firm has its own implied variance, realized variance, and therefore VRP. Bakshi & Kapadia (2003) and Carr & Wu (2009) document that the premium is present and negative (from the buyer's perspective) not only on index options but also on options written on individual equities, establishing the VRP at both the firm and market-level.

Furthermore, the VRP is related to broader themes in financial economics regarding how markets price risk. While the equity risk premium compensates investors for bearing systematic risk, the VRP specifically compensates for bearing variance risk. This distinction is important because volatility risk represents a second-order risk: it captures the magnitude and unpredictability of market fluctuations and tends to become particularly pronounced during periods of market stress (Drechsler & Yaron, 2011). The VRP thus sits at the intersection of option pricing theory, which provides the tools for extracting market-implied expectations, and asset pricing theory, which seeks to understand how risk is accounted for and priced.

It is worth noting that neither leg of this difference is observable. The risk-neutral expectation must be extracted from option prices, requiring either a parametric model or a model-free integration over a discrete and finite set of strikes. This study reads it from an interpolated volatility surface rather than from a single traded option contract. The physical expectation is not observable at all and must be replaced by a forecast, here the trailing realized variance under an assumption of persistence. The VRP is thus estimated as the difference between two proxies, and the measurement error in each transfers into all empirical results.

2.1.2 Early and Seminal Studies

Research on the VRP relies on methodological advance in the measurement of risk-neutral variance. Britten-Jones & Neuberger (2000) found that the expected variance under the risk-neutral measure can be extracted from a portfolio of out-of-the-money (OTM) options without assuming a parametric model on the underlying returns. This approach circumvents the constant volatility assumption in the Black-Scholes model. Jiang & Tian (2005) extend this and highlight its practical implementation, establishing it as the standard method for calculating implied variance measured used in option-pricing research.

The seminal contribution to the VRP literature is the study by Bollerslev et al. (2009). Constructing the VRP from high-frequency intraday returns on the S&P500, the authors show that the VRP contains significant information for future aggregate stock market returns, with an R^2 of 7 to 8 percent at the quarterly horizon. They interpreted this as evidence for the VRP capturing time-varying risk aversion, where periods of high VRP correspond to states in which investors demand greater compensation for bearing volatility risk. Bollerslev et al. (2014) then stress-test these findings for finite-sample bias. Through Monte Carlo simulations and a U.S. sample, they show that finite-sample biases in overlapping predictive regressions cannot account for the observed VRP predictability, and they replicate the predictive pattern in an international panel of eight equity markets. Finding that the predictive relationship is strongest when a global, capitalization-weighted VRP is used, suggesting that variance risk is priced as a common factor.

At the single-stock level Bakshi and Kapadia (2003) provided early evidence that volatility risk is priced in U.S. equity markets. Through delta-hedged option portfolios that isolate the exposure to volatility risk from directional market risk, they document significant negative average returns on these portfolios. This finding implies that investors are willing to accept a lower expected return in exchange for the protection that long-volatility positions provide, which is consistent with volatility risk carrying a negative market price. Their work laid the groundwork for understanding why the VRP is typically positive: resulting in sellers of variance being compensated for providing insurance against volatility shocks. Carr & Wu (2009) extend this to a broader cross-section of U.S. equity indices and individual stocks, documenting a persistently negative VRP (from buyer's perspective). Although the premium is systematically larger for index options than for single-stock options, reflecting additional compensation for index-level crash risk, it remains persistent at the firm level, indicating that the excess of implied over realized variance is not only an index-driven byproduct.

2.1.3 Theoretical Explanations for the VRP

The literature offers two potential explanations for the existence of the VRP, as reviewed by Zhou (2018). Firstly, the risk compensation hypothesis suggests that the VRP represents a reward for bearing crash or jump risk. Drechsler & Yaron (2011) take this intuition in a long-run risks model in which Epstein-Zin preferences combined with time-varying economic uncertainty imply that changes in consumption volatility are priced in. Demonstrating that agents demand a variance premium as compensation. Carr & Wu (2009) provide complementary research, showing that the spread between implied and realized variance loads primarily on jump risk. Under this view the premium is simply an equilibrium price, and a variance seller earns it as the rational reward for supplying insurance.

A distinct point, which should not be confused with the economic mechanism, is that part of the measured spread is mechanical. Model-free implied variance integrates option prices across strikes with a $1/K^2$ kernel, which assigns disproportionate weight to OTM puts, the contracts whose prices are inherently most sensitive to left-tail jump risks. Realized variance, reconstructed from the diffusive path of daily returns, cannot replicate that tail weighting. The two legs of the premium are as a result not measuring the same thing, and a portion of the difference between them reflects construction of the estimators rather than an outright economic reason.

A growing body of work further sharpens the role of jumps in shaping the VRP. Todorov (2010) uses high-frequency data to separate continuous stochastic volatility from discontinuous price jumps and shows that large jumps are almost always accompanied by immediate jumps in stochastic volatility. While the direct effect of a price jump on future realized volatility is found to mean-revert within half a day, the effect on the VRP is much more persistent: the premium spikes sharply after jumps and reverts to its mean only slowly. This suggests that investor risk aversion is time-varying and state-dependent, with risk tolerance dropping abruptly after adverse events even as mechanical volatility normalizes. Andersen et al. (2020) push this further by identifying a separate left-tail jump-intensity factor in international option panels. They find that in the post-crisis period, diffusive volatility normalizes while option-implied tail fear remains elevated for extended periods, and that it is just the tail factor, not diffusive volatility, that predicts future equity returns. Li & Zinna (2018) extract comparable factors from variance swap rates across multiple maturities and show that at the one month-horizon the jump component accounts for roughly 79 percent of the total VRP, and only 27

percent at the two-year horizon, consistent with the VRP being dominated by the pricing of short-term crash risk.

Another strand of the literature operationalizes the option market's view on future variance not by comparing implied to realized variances, but to a long-run historical inference. Stein (1989) documents that implied volatility itself is mean-reverting, and Goyal & Saretto (2009) sort the cross-section of single-name options on the spread between long-window historical and current implied volatility, finding strong predictive power for option returns. Cao & Han (2013) find the same for idiosyncratic volatility. While the long-baseline signal differs from the canonical VRP in both construction and interpretation (converging only when realized variance is close to its long-run mean), both approaches ultimately attempt to find whether option-implied variance is over or underpriced. This study therefore uses the canonical VRP as the main predictor, including the long-baseline approach as a robustness check.

The second perspective attributes the VRP to demand pressure and limits to arbitrage in option markets. Under this interpretation, the spread may partly come from systematic overpricing of options due to demand for hedges, limits to arbitrage, or behavioral biases such as loss aversion and probability weighting that cause investors to act irrationally and overestimate the likelihood of extreme events (Zhou, 2018). Bollen & Whaley (2004) document that net buying pressure from end users pushes index-option implied volatilities away from rational no-arbitrage levels, with the effect most present in the OTM segment. Garleanu et al. (2009) formalize this mechanism in a model in which option market-makers are risk-averse and cannot costlessly hedge, so that persistent demand imbalances translate into persistent price deviations. If the VRP comes from mispricing rather than being purely risk compensation, this would imply that strategies designed to capture this premium could generate persistent abnormal returns even after controlling for standard risk factors.

2.1.4 Cross-Sectional and Predictive Evidence

Beyond its role as a time-series predictor of aggregate equity returns, subsequent research has shown that the VRP also contains information for the cross-section of stock returns. Bali & Zhou (2016) outline that the VRP helps explain returns differences across portfolios sorted on various characteristics, even after controlling for traditional factors such as market beta, size, and momentum. They find that the VRP captures a source of systematic risk that is not fully accounted for by existing factor models. Cao & Han (2013) connect this to limits-to-arbitrage explanations through a cross-sectional dispersion in delta-hedged option returns being systematically related to the underlying stock's idiosyncratic volatility, with

hedging costs rising in volatility. Goyal & Saretto (2009) document a similar pattern: stocks for which historical realized volatility is high relative to option-implied volatility subsequently earn significantly higher straddle and delta-hedged option returns than stocks for which the two are close. DeMiguel et al. (2013) integrate this evidence into a portfolio-choice methodology, showing that using both option-implied skewness and the stock-level VRP in a mean-variance framework meaningfully improves out-of-sample Sharpe ratios, with the predictive power coming from the cross-sectional variation in these forward-looking option metrics rather than from their aggregate level.

Further refining the VRP signal, Feunou et al. (2018) decompose the total VRP into upside and downside components using corridor implied variance and signed realized variances. They find that the downside component is strongly positive and statistically drives almost fully the predictive power of the aggregate VRP, whereas the upside component is small, typically negative, and statistically insignificant. Pyun (2019) offers a methodological advance by estimating the VRP's predictive slope through a contemporaneous beta between market returns and variance innovations which deliver an out-of-sample R^2 of 6 to 8 percent at the monthly horizon, outperforming previously used rolling predictive regressions.

2.1.5 Measuring the Variance Risk Premium: Challenges and Implications

Nonetheless, the measurement of the VRP is not without its challenges. Jiang & Tian (2005) highlight that the construction of model-free implied volatility can be sensitive to the moneyness and time-to-maturity of the options used in its construction, making the empirical results sensitive to methodological choices. Gagliardini et al. (2016) show in a large-panel econometric setting that aggregating individual stocks into characteristic-sorted portfolios removes substantial cross-sectional information about risk premia, a finding that applied directly to VRP research. Their analysis motivates the use of individual stock panels with conditioning information over standard portfolio sorts whenever a study analyses the full cross-sectional variation in a priced factor.

The first difficulty concerns the risk-neutral leg. Britten-Jones & Neuberger (2000) show that expected variance under the risk-neutral measure can be taken from a sequence of OTM option prices without a parametric model of the underlying stock. Real option markets, however, supply a discrete and finite set of strikes. Jiang & Tian (2005) demonstrate that the truncation and discretization errors are as a result material, and that the estimate is sensitive to the moneyness, and time to maturity of the contracts. This emphasizes how sensitive the results are to measurement choices: two studies of the same underlying report different premia simply

because of their strike-selection rules. The $1/K^2$ weighting discussed in Section 2.1.3 compounds this, as it focuses the estimate on exactly the deep OTM segment where quotes are the thinnest and bid-ask spreads the widest. This thesis avoids the strike-selection problem by taking implied variance from the OptionMetrics 30-day standardized surface on a fixed delta grid (Section 3.2.1). This keeps moneyness and maturity constant across firms, but the estimate then depends on the interpolation used to construct the surface.

The second difficulty concerns the physical leg. The object required is a conditional expectation of future variance; what is available is a realization. Replacing a trailing realized variance introduces measurement error into the predictor, drawing estimated slopes towards zero, and forcing a persistence assumption that holds well in calm markets and poorly after a jump (Todorov, 2010). Other approaches substitute one problem for another: high-frequency realized variance reduces sampling error at the cost of microstructure noise, and long-baseline historical volatility, similarly to Goyal & Saretto (2009) is more stable but no longer aligned with the horizon of the option. This thesis uses the trailing 21-day realized variance as the physical leg (Section 3.2.2), and takes the long-baseline alternative only as a robustness check (Section 5.6).

2.2 Short-Variance Strategies

2.2.1 Definition and Mechanics

A short-variance strategy sells exposure to future variance, collecting a payoff equal to the difference between the implied variance priced into options at the time of entry, and the realized variance that subsequently materializes. The payoff of a single period short variance swap with unit notional can be written as:

$$Payoff(i, t \rightarrow t+1) = IV^2(i, t) - RV(i, t, t+1)$$

where $IV^2(i, t)$ is the risk-neutral expectation of variance for asset i at time t, and $RV(i, t, t+1)$ is the subsequent realized variance. Because implied variance typically exceeds subsequent realized variance, the short-variance position generates a positive payoff in most periods. This payoff structure makes short-variance strategies a natural vehicle for harvesting the VRP (Carr & Wu, 2009).

In practice, short-variance exposure can be obtained through several instruments, including over-the-counter variance swaps, listed volatility swaps, synthetic variance-swap positions constructed from static portfolios of options, and delta-hedged short option positions.

Bondarenko (2014) evaluated the replication of discretely sampled variance swap using a static basket of options combined with a dynamic trading strategy in the underlying, extending earlier continuous-time replication arguments to settings including jumps and discrete rebalancing. He demonstrates that the resulting payoff reproduces the economics of a variance swap without requiring frictionless continuous time trading, but that deviations from the theoretical rebalancing schedule generate substantial hedging errors, a relevant point when evaluating the statistical properties of strategy returns. At the institutional level, covered-call strategies, in which investors hold a long equity position and write call options against it, embed a short-volatility exposure that can be analytically decomposed into a delta-related equity component and a residual volatility-selling component (Israelov & Nielsen, 2015).

2.2.2 Short-Variance Returns: Empirical Evidence

Empirical evidence overwhelmingly supports the observation that implied volatility tends to exceed realized volatility on average, generating a persistent premium for sellers of variance. Bondarenko (2014) quantifies this at the index level, estimating an unconditional VRP of roughly -26.3 percent monthly for the S&P500 over a 20-year period. He estimates that only a limited portion of the premium is explained by standard equity risk factors, pointing to a distinct pricing of volatility risk in index options.

Further evidence from the study of covered call strategies reinforces these findings. Hill et al. (2006) and Israelov & Nielsen (2015) document that buy-write strategies (combining long equity position and a short call option, embedding short-volatility exposure) on the S&P500 have, in certain sample periods, outperformed their underlying benchmarks on a risk-adjusted basis. The authors infer that the outperformance is attributable to the persistent sale of overpriced implied volatility rather than to directional equity exposure, that is, the return the position earns simply from being long the underlying equity. Israelov & Nielsen (2015) are careful to separate the buy-write into a delta-related equity component and a residual volatility-selling component, demonstrating that the latter contributes to a major share of the outperformance. However, they also note that realized outperformance is concentrated in low-volatility regimes and can reverse sharply during volatility spikes.

The risk profile of short-variance positions merits emphasis, because it has direct implications for how the empirical results should be interpreted. The payoff distribution of short-variance strategies is particularly negatively skewed and leptokurtic. They tend to generate small, frequent gains during periods of calm markets, but are susceptible to large losses during volatility spikes or market crashes. This asymmetric payoff profile is comparable to

selling insurance, where the seller collects steady premiums but can face catastrophic payouts in extreme circumstances (Eraker, 2021). At the single-stock level this asymmetry is amplified, as the individual stocks are exposed to idiosyncratic jumps (earnings surprises, credit events). Todorov (2010) shows that after a large price jump, the VRP spikes and remains elevated long after mechanical volatility has normalized, translating directly into losses for variance sellers entering positions in the immediate aftermath of such events. Standard risk measures such as Sharpe ratios, and linear-factor alphas estimated over short windows, therefore tend to overstate the attractiveness of short-variance payoffs.

2.3 The Relationship between the VRP and Short-Variance Strategy Returns

2.3.1 Predictive Power of the VRP

The economic reasoning linking the VRP to the return of a short-variance strategy is direct. When the VRP is high, implied variance is rich relative to the variance that is likely to materialize. The option market is pricing in more variance than the physical expectation warrants, reflecting elevated compensation for idiosyncratic volatility risk. The payoff measured over the subsequent window is, by construction, the realized difference between implied and realized variance, so a higher ex ante VRP implies a higher expected payoff.

However, at the single stock level the same mechanism is expected to operate but with noisier realizations. A high VRP does not appear arbitrarily: implied volatility is often elevated precisely because there are distinct reasons for the market to expect turbulence. For individual firms, these drivers mainly include idiosyncratic jumps and market-wide shocks. In such states, an elevated VRP may partly reflect a rational upward adjustment in the probability of high realized variance, over simply pure overpricing. If the anticipated event materializes, the short-variance strategy will most likely suffer significant losses regardless of how attractive the ex-ante appeal of the VRP signal. The predictive power of the VRP for variance selling payoffs therefore depends on whether, on average and in the cross-section, elevated implied variance systematically overstates subsequent realized variance. The cross-sectional evidence of Cao & Han (2013), Goyal & Saretto (2009), and DeMiguel et al. (2013) suggests that on average it does, while Todorov (2010) and Feunou et al. (2018) emphasize that the predictability is concentrated in the component of the VRP associated with downside and jump risk, reinforcing the importance of distinguishing signal from noise.

2.3.2 Risk-Adjusted Performance and the Fama-French Framework

A critical question in evaluating returns of short-variance strategies is whether observed profits represent genuine abnormal performance (alpha) or simply compensation for exposure to systematic risk factors. The Fama-French five-factor model (Fama & French, 2015) serves as the appropriate benchmark for this assessment. The five factors: market excess return (MKT), size (SMB), value (HML), profitability (RMW), and investment (CMA), represent the most comprehensive set of common risk factors in the current asset pricing literature. By regressing the excess returns of short-variance strategy on these factors, the intercept (alpha) captures any performance that cannot be attributed to exposure to these systematic risk sources.

A positive and significant alpha is typically interpreted as evidence that the VRP contains information about variance pricing that is not captured by standard equity risk factors. Supporting the interpretation that volatility markets exhibit exploitable patterns, consistent with the market-friction perspective. If, on the other hand, the alpha is insignificant after risk adjustment, the returns are better explained as compensation for bearing risks that are already captured by the Fama-French factors, which would be more consistent with the risk compensation hypothesis. The existing evidence on which of the two perspectives dominates is mixed. Bondarenko (2014) finds that the large unconditional VRP on the S&P 500 cannot be explained by size, value or momentum factors, and Bali & Zhou (2016) show that VRP-related measures add explanatory power even after controlling for market beta, size and momentum. On the other hand, Todorov (2010) and Andersen et al. (2020) argue that VRP predictability reflect risks that are not captured by standard diffusion-based models. This suggests that linear factor models built on continuous dynamics may not fully address the relevant sources of risk.

Alongside this interpretive ambiguity, the statistical properties of short-variance payoffs create constraints on inference. Variance-related time series exhibit pronounced persistence and clustering: periods of high volatility tend to be followed by further high volatility (Engle & Bollerslev, 1986), and the payoffs of short-variance strategies inherit this serial dependence. Without adjustment, the standard errors associated with regression-based performance tests will be understated, leading to overly optimistic inferences about abnormal performance.

2.3.3 Research Expectations

Building on the literature, this thesis expects a positive relationship between the stock-level VRP and the abnormal returns of short-variance strategies. First, it is hypothesized that stocks showing higher option-implied variance relative to their expected realized variance will systematically generate higher payoffs for variance sellers, leading to a positive and statistically

significant panel beta on the VRP. Second, as prior research emphasizes that predictability is often concentrated in the components associated with downside and jump risk, this study expects the downside leg of the premium to primarily drive this relationship. Third, if the downside leg reflects a structural feature of variance pricing rather than a feature of a single period, its coefficient is expected to stay constant across multiple sample sub-periods.

Fourth, to confirm whether these statistical signals contain information about variance pricing. This should translate in the returns of a strategy trading on it. A long-short strategy that sells variance in the stocks with the highest VRP and buys variance with the lowest VRP is therefore expected to generate a positive alpha after controlling for the Fama-French five factors, momentum and changes in aggregate volatility (VIX). A positive alpha indicates that the firm-level VRP captures pricing information not fully accounted for by conventional equity risk factors.

3 Data

3.1 Sample

The empirical analysis is conducted on a panel of seventy-three large-capitalisation U.S. equities observed at the daily frequency between 1 January 1996 and 30 November 2024. The universe is not drawn from an index, but is a hand-picked list of firms across sectors, to which ten firms that either failed, were acquired under distress, or were otherwise severely impaired during the sample period (Enron, WorldCom, Lehman Brothers, Wachovia, AIG, Palm, Citigroup, Bank of America, General Electric and American Express) were deliberately added. Each ticker is resolved to a single CRSP security (PERMNO) using the history of that ticker in crsp.stocknames, restricted to ordinary shares; where a ticker was used by more than one firm during the sample window, the firm with the largest average market capitalisation while carrying the ticker is retained, and that firm's full history is used. OptionMetrics security identifiers are then linked to these PERMNOs through the OptionMetrics-CRSP link table wrdsapps.opcrsphist, using only the dates for which each link is valid. All seventy-three firms are matched, and the final, unbalanced estimation panel comprises 22,858 firm-month observations.

The ten distressed firms account for 10.1 percent of all firm-months, but for 25.3 percent of the worst one percent of firm-month payoffs. This proportion is deliberately high relative to their weight in the market. Given the hypotheses and methodology used, which focus on

downside risk, stability across periods and the losses of a short-variance strategy, these observations are likely to be the focal points of the analysis. For this reason the results are also reported for the universe without them (Section 5.6). Three of the ten firms, Enron, WorldCom and Lehman Brothers, went bankrupt during the sample period. The collapses of Enron and Lehman Brothers fall inside the panel, but that of WorldCom does not: CRSP reports no returns for WorldCom between 26 and 28 June 2002, so no complete forward window covers its fall in early July 2002. The sample should therefore be seen as a set of prominent firms in which distress episodes are over-represented on purpose, rather than as a formally representative or survivorship-bias-free cross-section of U.S. equities.

Data is drawn from five sources. Option-implied variances are constructed from the OptionMetrics IvyDB U.S. standardized volatility surface tables `vsurfd{YYYY}`, accessed yearly between 1996 and 2024. Daily equity returns and shares outstanding are obtained from CRSP. Asset-pricing factors, such as the Fama and French (2015) five factors and the Carhart (1997) momentum factor, are taken from `ff_all.fivefactors_daily` and `ff_all.factors_daily`. The CBOE volatility index used as a macro variance control is obtained from `cboe_all.cboe`. The term-spread and default-spread macro controls (further discussed in Section 4.6) are pulled from the `frb.rates_daily` release on WRDS. All time-series are aligned on calendar dates and aggregated to month-end before merging. All data used is summarized in Appendix A.

The implied volatilities, daily returns, factor returns, VIX and interest rates are downloaded directly from WRDS, and every variance measure, premium and payoff in Section 3.2 is computed from these in this thesis. The OptionMetrics implied volatilities are already annualized, so squaring them gives an annualized variance, while the realized variances are annualized here by the factor 252/21. Of the 483,223 CRSP stock-days and 479,469 firm-days with a usable put and call wing, 476,698 firm-days have a complete set of implied and realized measures. From these, the last firm-day of each month with complete data is kept, which is the last trading day of the month for 99.6 percent of firm-months. No firm is lost in the matching, no month has no observations. A small number of extreme values (for example two firm-months with an implied volatility above 200 percent) are kept and handled by the winsorization to the VRP and payoff series described in Section 4.6, rather than removed.

3.2 Variables

3.2.1 Implied Variance Measures

For each firm *i* on each trading day *t*, the OptionMetrics IvyDB U.S. standardized volatility surface provides implied volatilities at fixed maturities and option deltas. These observations are constructed rather than raw quotes from individual option contracts. OptionMetrics starts with the cross-section of listed options available for firm *i* on trading day *t*, each with its own strike, expiry, bid-ask quote and Black-Scholes implied volatility, and then projects this information onto a standardized grid of maturities and deltas. The projection uses smoothing across the maturity and delta dimensions, so that contracts closer to the target maturity and delta receive more weight (Aït-Sahalia & Lo, 1998; OptionMetrics, 2023). The thirty-calendar-day slice used in this thesis is therefore an interpolated surface point rather than a single listed contract. This standardization makes implied volatilities comparable across firms and trading days, even though the underlying option contracts differ.

From this, risk-neutral variance measures are computed by trapezoidal integration of squared implied volatility across the surface delta grid. Specifically, the implied variance is obtained by integrating squared implied volatility across the standardized delta grid by the composite trapezoidal rule. For ordered grid points $\delta_0 < \delta_1 < \ldots < \delta_K$ where δ denotes the absolute delta |Δ|, with values $y_k = IV(\delta_k)^2$, the integral $\int y \, d\delta$ is approximated by the sum over k = 1, …, K:

$$\sum_k \frac{1}{2}(y_{k-1} + y_k)(\delta_k - \delta_{k-1})$$

so that each interior delta node receives full weight and the two boundary nodes (|Δ| = 10 and |Δ| = 90 for the total measure, |Δ| = 10 and |Δ| = 50 for the downside and upside legs) receive half weight. The integral is then normalized by the corridor width (δ_max − δ_min) to yield a delta-weighted average squared implied volatility. The process is applied separately to the put and call wing, with the total measure defined as their average. This is the delta-domain analogue of the model-free implied variance of Britten-Jones and Neuberger (2000), in the corridor method used by Andersen & Bondarenko (2007) and Feunou et al. (2018).

- $IV_var_total_{i,t}$ — Total option-implied variance, computed as the average of the put-wing and call-wing integrals over absolute deltas in [10, 90].

- *IV_var_down*$_{i,t}$ — Downside (put-wing) implied variance, restricted to OTM puts with absolute deltas in [10, 50].
- *IV_var_up*$_{i,t}$ — Upside (call-wing) implied variance, restricted to OTM calls with absolute deltas in [10, 50].

The cut |Δ| = 50 follows the corridor implied variance approach of Andersen & Bondarenko (2007), Bondarenko (2014) and Gonzalez-Perez (2015), and adapts the model-free decomposition of Feunou et al. (2018) from the index level to the fixed-delta firm-level surface. As both selected ranges (absolute deltas in [10, 50] on the put wing and on the call wing) end at the at-the-money point (|Δ| = 50), they include the most liquid, high-gamma segment of the options market. As a result, these variance measures capture asymmetry in and around the at-the-money (ATM) segment, rather than deep in the tails. While excluding this central section would better isolate extreme left-tail jump risk, doing so at the firm level sacrifices too much measurement precision. A tail-only version of the signal, restricted to |Δ| in [10, 25] on both wings, is therefore used only as a robustness check (Section 5.6). All variance measures are annualized in decimal units: the OptionMetrics implied volatilities are already annualized, so their squares are annualized variances.

3.2.2 Realized Variance Measures

Realized variance and realized semi-variances are constructed from CRSP daily return series following Barndorff-Nielsen et al. (2008). The notation *RS_back_down*$_{i,t}$ and *RS_back_up*$_{i,t}$ denote the downside and upside realized semivariances. Let d denote the last trading day of the month t with complete data for firm i, and let τ index trading days. For a window of w = 21 trading days:

$$RV_back_{i,t} = (252/w) \cdot \Sigma_{\tau \in [d-w+1,\, d]}\, r_{i,\tau}^2$$

$$RS_back_down_{i,t} = (252/w) \cdot \Sigma_{\tau \in [d-w+1,\, d]}\, r_{i,\tau}^2 \cdot \mathbb{1}\{r_{i,\tau} < 0\}$$

$$RS_back_up_{i,t} = (252/w) \cdot \Sigma_{\tau \in [d-w+1,\, d]}\, r_{i,\tau}^2 \cdot \mathbb{1}\{r_{i,\tau} > 0\}$$

where $\mathbb{1}\{\cdot\}$ is the indicator function, with corresponding forward-looking analogues *RV_fwd*$_{i,t+1}$, *RS_fwd_down*$_{i,t+1}$, and *RS_fwd_up*$_{i,t+1}$ computed over the next twenty-one trading days, τ ∈ [d+1, d+21], matching the implied-variance slice.

The factor 252/21 annualizes the 21-day variance, so all realized and implied variance share the same annualized-variance units.

3.2.3 Variance Risk Premia

Three stock-level VRP measures are defined as the difference at time t between the appropriate risk-neutral and physical variance proxies, using the trailing realized variance as the conditional variance forecast:

$$VRP_total_{i,t} = IV_var_total_{i,t} - RV_back_{i,t}$$
$$VRP_down_{i,t} = IV_var_down_{i,t} - RS_back_down_{i,t}$$
$$VRP_up_{i,t} = IV_var_up_{i,t} - RS_back_up_{i,t}$$

All three measures are annualized in variance units. The construction follows the specification of Bollerslev et al. (2009) at the index level and the downside and upside legs follow the idea of Feunou et al. (2018). Unlike their decomposition, the two legs do not add up to VRP_total: each implied leg is the average squared implied volatility over one wing and, as a result, has the same scale as IV_var_total, while the two realized semivariances add up to RV_back. VRP_down and VRP_up are therefore taken as separate signals rather than as components of the VRP_total construction.

Note that the VRP is an ex-ante signal: every component is dated at, or before, the close of day d, the last trading day of month t with complete data. The realized variance term $RV_back_{i,t}$ is the trailing twenty-one-day realized variance, used here as a model-free proxy for the conditional expectation $E_t[RV_{i,t\to t+21d}]$ under the persistence assumption of Bollerslev et al. (2009).

3.2.4 Short-Variance Strategy Payoffs

The realized payoff to a one-period short-variance trade initiated at the close of day d of month t and unwound twenty-one trading days later is defined as the difference between the implied variance entered at t and the variance that materializes over the following window:

$$payoff_{i,t+1} = IV_var_total_{i,t} - RV_fwd_{i,t+1}$$
$$payoff_d_{i,t+1} = IV_var_down_{i,t} - RS_fwd_down_{i,t+1}$$
$$payoff_u_{i,t+1} = IV_var_up_{i,t} - RS_fwd_up_{i,t+1}$$

The payoff is reported in the same annualized variance units as the inputs. A positive payoff corresponds to a profit for the variance seller and is the dependent variable in all

predictive regressions. A high VRP signal at t yields a profitable short-variance trade only when the resulting payoff is positive, that is when realized variance over the holding period falls below the implied variance at initiation. In percentage terms, dividing the payoff by the implied variance at portfolio formation, $r^{VS}_{i,t+1} = payoff_{i,t+1} / IV_var_total_{i,t}$, yields the realized return on a unit-notional variance-swap position (Carr & Wu, 2009; Bondarenko, 2014). This conversion is used for the trading strategy of Section 4.4, so that the Jensen alpha of equation (4) can be interpreted as an abnormal percentage return rather than an abnormal annualized variance.

3.3 Control Variables

This study controls for systematic exposures to standard equity factors. Specifically, the Fama & French (2015) five factors: the excess market return *mktrf*, the size factor *smb*, the value factor *hml*, the profitability factor *rmw*, and the investment factor *cma.* These are augmented by the Carhart (1997) momentum factor *umd* and the first difference of the CBOE volatility index (*dvix*). The latter is included as a parsimonious proxy for changes in aggregate variance risk and follows the use of changes in the VIX as a market-wide variance factor. All factors are aggregated from daily to monthly frequency by compounding daily returns within calendar months (*dvix* is summed within the month). The lagged level of the VIX index, the term spread (10-year minus 3-year Treasury yield), and the default spread (Moody's Baa minus Aaa corporate bond-yield), retrieved from `frb.rates_daily`, are also used as macro-state controls.

3.4 Descriptive statistics

Tables 1 and 2 present the summary statistics and correlation matrix for the variance and payoff metrics, with the VRP and payoff series winsorized at the 1st and 99th percentile within each calendar month. A glance at the metrics immediately reveals the pronounced non-normal distribution and heavy-tailed nature of the data. Most notably, the aggregate series demonstrate a pronounced difference between their means and medians. While the total firm-level VRP (*vrp_total*) and the corresponding short-variance payoff (*payoff*) have negative sample means (−0.004 and −0.008) their medians are positive (0.015 and 0.015). This dynamic captures the asymmetric risk profile of volatility selling: frequent, modest gains with large drawdowns (as shown by the extreme minima of −16.38 for *vrp_total* and −22.38 for *payoff*). These drawdowns are shown by the extreme spikes in underlying volatility. With the 21-day

realized variance *rv_back_21d* for instance, sitting at 0.121 in the 75th percentile, but with a maximum of 26.54. Finally, the Pearson correlation between *vrp_total* and *payoff* is 0.149, between *vrp_down* and *payoff* is −0.028, and between *vrp_up* and *payoff* is 0.025.

Table 1. Descriptive statistics of variance and short-variance variables.

Variable	Mean	SD	Min	25%	Median	75%	Max
Panel A. Implied variance							
Implied Variance Total *(iv_var_total)*	0.121	0.162	0.010	0.047	0.077	0.134	4.625
Implied Variance Down *(iv_var_down)*	0.132	0.180	0.001	0.050	0.083	0.147	4.866
Implied Variance Up *(iv_var_up)*	0.106	0.150	0.006	0.040	0.067	0.119	8.056
Panel B. Realised variance							
Realised Variance (21d) *(rv_back_21d)*	0.127	0.354	0.001	0.030	0.059	0.121	26.537
Realised Semivariance Down *(rs_back_down)*	0.058	0.168	0.000	0.011	0.024	0.055	11.168
Realised Semivariance Up *(rs_back_up)*	0.070	0.218	0.000	0.015	0.029	0.063	15.369
Panel C. Variance risk premia							
VRP Total *(vrp_total)*	−0.004	0.215	−16.381	−0.007	0.015	0.035	1.755
VRP Down *(vrp_down)*	0.074	0.118	−6.926	0.030	0.053	0.093	2.009
VRP Up *(vrp_up)*	0.037	0.130	−7.391	0.012	0.029	0.059	3.285
Panel D. Short-variance payoffs							
Payoff Total *(payoff)*	−0.008	0.294	−22.384	−0.007	0.015	0.037	1.870
Payoff Down *(payoff_d)*	0.072	0.185	−12.919	0.027	0.051	0.095	2.760
Payoff Up *(payoff_u)*	0.036	0.130	−8.231	0.014	0.030	0.058	3.355

Notes. The table reports pooled firm-month summary statistics for all variables. Sample: 22,858 firm-month observations on 73 large-capitalization U.S. equities, January 1996 – November 2024. Winsorized at the 1st and 99th percentile within calendar month for VRP and payoff series.

Table 2. Pearson correlation matrix of variance-risk-premium signals and the short-variance payoff.

Variable	(1)	(2)	(3)	(4)
(1) VRP Total *(vrp_total)*	1.000	0.451	0.741	0.149
(2) VRP Down *(vrp_down)*	0.451	1.000	0.324	−0.028
(3) VRP Up *(vrp_up)*	0.741	0.324	1.000	0.025
(4) Payoff Total *(payoff)*	0.149	−0.028	0.025	1.000

Notes. The table reports pairwise Pearson correlation coefficients across the variables defined in Sections 3.2.3 and 3.2.4 over the pooled firm-month sample (N = 22,858; January 1996 – November 2024).

4 Method

4.1 Estimation Strategy

The empirical strategy primarily uses a panel predictive regression, supported by Fama-MacBeth regressions, calendar-time portfolio sorts, and out-of-sample forecasting tests detailed in Sections 4.4 and 4.5. Each expectation is tested with a specific empirical specification, as shown below. Following Petersen (2009) and the VRP literature (Bollerslev et al., 2009, 2014), inference in the main specification uses one-way standard errors clustered by time, to account for strong cross-sectional dependence that occurs when changes to aggregate variance affect many firms simultaneously.

4.2 Panel Predictive Regressions – H_1 and H_2

The first research expectation (H_1) is that the firm-level total VRP positively predicts the next-month short-variance payoff: firms whose implied variance is high relative to their trailing realized variance should, on average, earn a higher payoff from selling variance over the following month. In equation (1) below, this corresponds to a positive beta on VRP_total:

$$payoff_{i,t+1} = \alpha + \beta \cdot VRP_total_{i,t} + \varepsilon_{i,t+1} \qquad (1)$$

where, restating from Section 3.2:

$$payoff_{i,t+1} = IV_var_total_{i,t} - RV_fwd_{i,t+1} \quad \text{(in variance units)}$$

$$VRP_total_{i,t} = IV_var_total_{i,t} - RV_back_{i,t} \quad \text{(in variance units)}$$

$IV_var_total_{i,t}$ (by trapezoidal rule on the OptionMetrics 30-day delta grid)

$RV_back_{i,t} = (252/21) \cdot \Sigma_{\tau \in [d-20,\, d]}\, r_{i,\tau}^2$ (the trailing 21-day annualized realized variance)

$RV_fwd_{i,t+1} = (252/21) \cdot \Sigma_{\tau \in [d+1,\, d+21]}\, r_{i,\tau}^2$ (the forward 21-day annualized realized variance)

d is the last trading day of month t with complete data for firm i, and τ indexes trading days.

Equation (1) is estimated by ordinary least squares (OLS) with standard errors clustered by month. The null hypothesis associated with H_1 is $\beta = 0$, with a $\beta > 0$ alternative.

The second expectation (H_2) is that the downside leg of the VRP primarily drives the predictive content of the total VRP, similar to the decomposition of Feunou et al. (2018). This is tested with a horse race: both premia enter the same regression, so that each coefficient measures the predictive content of one premium. If the downside leg carries the information, its coefficient stays positive and significant while the coefficient on the total premium falls towards zero; if the total premium carries information beyond the downside leg, the reverse holds. The horse-race specification is then:

$$payoff_{i,t+1} = \alpha + \beta_{tot} \cdot VRP_total_{i,t} + \beta_{dn} \cdot VRP_down_{i,t} + \varepsilon_{i,t+1} \qquad (2)$$

The null hypothesis associated with H_2 is $\beta_{dn} = 0$, with a $\beta_{dn} > 0$ alternative. H_0 is rejected if the coefficient on VRP_down is positive and significant; if, in addition, the coefficient on VRP_total falls near zero, this would mirror the index-level downside protection interpretation. A significant coefficient on VRP_total once VRP_down is included implies instead that the total VRP contains additional predictive information. In addition to equation (2), a horse race substitutes $VRP_up_{i,t}$ for $VRP_total_{i,t}$ and tests whether the upside leg adds predictive power beyond the downside leg, extending the approach of Feunou et al. (2018) to the firm level. To complement the panel estimator, monthly cross-sectional regressions of $payoff_{i,t+1}$ on $VRP_total_{i,t}$ and on $VRP_down_{i,t}$ are estimated independently for each month, with the resulting time series of monthly slopes then summarized by its mean. The Fama & MacBeth (1973) regressions provide a cross-sectional robustness test of the results in equations (1) and (2). Significance is evaluated using Newey-West (1987) heteroskedasticity and autocorrelation consistent (HAC) t-statistics with three lags.

4.3 Subsample Estimation – H_3

The third expectation predicts that the coefficient on VRP_down, estimates with equation (1) with VRP_down in place of VRP_total, is constant across sample sub-periods, which would establish the downside leg as a structural feature of cross-sectional variance

pricing rather than a regime specific anomaly. Three sub-periods are defined: a pre-GFC (1996-2007), a GFC and post-GFC sample (2008-2015), and a modern sample (2016-2024). These periods are chosen to reflect distinct market environments rather than to isolate specific macro events. The pre-GFC era is characterized by predominantly institutional traders in the single-name option market, the GFC and post-GFC by clustered volatility shocks and post-crisis balance sheet constraints on market makers; and the modern era by the expansion of single-name option volumes and the emergence of retail and systematic variance strategies. The null hypothesis under H_3 is that the three sub-period coefficients on *VRP_down* are equal:

$$\beta_{dn,\,1996–2007} = \beta_{dn,\,2008–2015} = \beta_{dn,\,2016–2024} \qquad (3)$$

This is tested by pooling the three sub-periods, interacting VRP_down with dummies for the second and third sub-period, and testing with a Wald test to assess whether both interaction coefficients are jointly zero. Standard errors are clustered monthly. The coefficient estimates, t-statistics and R-squared values for each sub-period are also reported. The same procedure is then applied with VRP_total in place of VRP_down, with the analysis reported in Sections 5.3 and 5.6.

4.4 Trading Strategy and Risk-Adjusted Performance – H_4

The fourth expectation predicts that a long-short trading strategy that takes short-variance positions in firms with the highest VRP signals and buys variance in the firms with the lowest VRP signals, generates positive risk-adjusted alpha after controlling for the factors in equation (4). The strategy is implemented as a calendar-time long-short portfolio. At the close of each month *t*, firms are sorted into quintiles by $VRP_total_{i,t}$ and a portfolio is formed that is short variance in the top quintile (stocks with highest VRP), and long variance in the bottom quintile. Quintile sorts are taken over deciles due to the sample size; months in which fewer than fifteen firms are available, and in which fewer than five distinct VRP values are observed are excluded from the strategy. To express performance in interpretable percentage units, the realized payoff is converted into a variance-swap return by dividing it by the implied variance entered at portfolio formation, following Carr & Wu (2009) and Bondarenko (2014). The portfolio is held for the following month and rebalanced at month-end. The realized return of the portfolio in month t+1 is calculated as the equal-weighted average variance-swap return in the top quintile minus the equal-weighted average variance-swap return in the bottom

quintile. The risk-adjusted performance of the long-short portfolio is evaluated by the following time-series regression (4):

$$r_{LS,t} = \alpha + \beta_1 \cdot mktrf_t + \beta_2 \cdot smb_t + \beta_3 \cdot hml_t + \beta_4 \cdot rmw_t + \beta_5 \cdot cma_t + \beta_6 \cdot umd_t + \beta_7 \cdot \Delta VIX_t + \varepsilon_t \qquad (4)$$

where:

$r_{LS,t}$ = Monthly long-short variance-swap return (top minus bottom quintile) over holding month t, from a portfolio formed at the end of month t-1, in decimal units; all factors are measured over the same month t

$mktrf_t$, smb_t, hml_t, rmw_t, cma_t = Fama and French (2015) five factors

umd_t = Carhart (1997) momentum factor

ΔVIX_t = the first difference of the CBOE volatility index

α = Jensen's alpha

Equation (4) is estimated by OLS with Newey-West (HAC) standard errors at three lags, the null hypothesis under H_4 is $\alpha = 0$ and the alternative is $\alpha > 0$. A positive and statistically significant intercept implies that the stock-level VRP signal generates alpha that is not absorbed by exposure to the standard equity factors or changes in the VIX. For interpretation, monthly mean returns and alphas are annualized by multiplying by twelve, and volatility is annualized by multiplying by the square root of 12.

4.5 Out-Of-Sample Predictive Ability

The in-sample regressions in equations (1) and (2) are complemented by an out-of-sample (OOS) forecasting test following Welch & Goyal (2008). For each calendar month t after a training period ending in 1999, 2002 or 2005, equation (1) is re-estimated using all available prior observations and used to forecast next-month payoff for each firm. The forecasts are benchmarked against the historical mean of *payoff* over the same estimation window. The OOS R-squared is defined as:

$$R^2_{OOS} = 1 - \Sigma_{i,t} \left(payoff_{i,t+1} - \hat{payoff}_{i,t+1}\right)^2 / \Sigma_{i,t} \left(payoff_{i,t+1} - \bar{payoff}_t\right)^2 \qquad (5)$$

where the sums are taken across all firms and forecast months in the test period. $\hat{payoff}_{i,t+1}$ is the forecast obtained by applying equation (1), estimated on all observations dated

before month t, to $VRP_total_{i,t}$ and $\bar{payoff}_t$ is the mean of payoff over the same observations (the prevailing mean). A positive out-of-sample R-squared therefore indicates that forecasts based on VRP_total outperform the historical-mean benchmark. The same procedure is applied to VRP_rich using a year-end 2000 training cutoff. Statistical significance is assessed by the Clark & West (2007) one-sided test of equal predictive accuracy in nested models, with Newey-West HAC standard errors at three lags. As a sensitivity check, the exercise is repeated using training cutoffs of 1999, 2002 and 2005; a fourth cutoff at year-end 2015 starts the expanding window on 1996–2015 and evaluates the forecasts over the modern era only, 2016–2024.

4.6 Robustness Checks

The empirical findings are supplemented by a broad set of robustness tests. All analyses winsorize the VRP, payoff and variance-swap return series within each calendar month at the 1st and 99th percentile. Observations below or above these thresholds are replaced by the corresponding percentile value. Applying this month by month prevents the most unusually volatile months from setting the bounds for the full sample. Winsorization affected 3.0 percent of observations for each variable and 7.8 percent of firm-months on at least one variable. Because standard long-short strategies are often constructed untrimmed, the long-short alpha is also re-estimated under no winsorization and under a tighter $2.5^{th}/97.5^{th}$ percentile bounds. The panel standard errors on equation (1) are also tested with two-way clustering by firm and month (Cameron et al., 2011) and with Driscoll and Kraay (1998) HAC standard errors at three lags. A firm fixed-effects specification of equation (1) is also estimated to test whether the predictive relationship holds within firms. In addition, lagged VIX, first difference of the VIX, the term spread, and the default spread are added to test whether the predictability survives conditioning on standard macro predictors.

The subsample analysis is further refined by separating the crisis era into acute crisis (2008-2009), recovery (2010-2012), calm (2013-2015) and COVID (2020) periods. To distinguish calendar effects from market volatility conditions, months are classified by the VIX level at portfolio formation. High-VIX months are defined as the top-tercile months of the full-sample VIX distribution, with all the remaining months forming the control group. Equation (1) is re-estimated within each volatility regime, and an interaction specification is used to test whether the predictive slope differs between high- and lower-volatility periods:

$$payoff_{i,t+1} = \alpha + \beta_1 \cdot VRP_total_{i,t} + \beta_2 \cdot (VRP_total_{i,t} \times HighVIX_t) + \beta_3 \cdot HighVIX_t + \varepsilon_{i,t+1} \quad (6)$$

where:

$HighVIX_t$ = Dummy variable that equals 1 when closing VIX in month *t* lies in top tercile of the in-sample distribution, zero otherwise

The coefficient β_2 captures the additional predictive slope in high-volatility months over the baseline. A β_2 coefficient not statistically different from zero indicates that the equality across regimes cannot be rejected, although it does not prove that the relationship is identical.

The OOS analysis is supplemented by the Clark-West one-sided test, alternative training cutoffs at year-end 1999, 2002 and 2005, and by the modern-regime cutoff at year-end 2015, using 1996-2015 as the initial estimation window. The trading results are also evaluated under a transaction-cost overlay of 0, 10 and 25 basis points per leg. Together with higher-moment and tail-risk diagnostics for the long-short portfolio, including skewness, excess kurtosis, Sortino ratio, five percent expected shortfall and cumulative return peak-to-trough drawdown. Finally, the robustness program includes the alternative variance-swap signal VRP_rich = IV_var_total − HV_long252d in the spirit of Goyal & Saretto (2009), a tail-only version of VRP_total built from the $|\Delta| \in [10, 25]$ wings of the surface, a re-estimation excluding the ten distressed firms and the 2008–2009 crisis. Full results for these tests are reported in the following chapter.

5 Results

5.1 The predictive content of the total variance risk premium (H_1)

The first research expectation predicts that the firm-level total VRP exhibits a positive coefficient on next-period short-variance payoffs. Column (1) of Table 3 reports the estimate of equation (1). The slope on VRP_total is positive and strongly significant, with a coefficient of 0.204 (t = 5.60). The corresponding R^2 is only 2.2%, so while the relationship is statistically strong, most of the variation in firm-level payoffs is left unexplained. The cross-sectional Fama-MacBeth estimator in Model 1 of Table 4 corroborates this finding, where the average monthly slope on VRP_total is 0.141 with a Newey-West t-statistic of 2.16. Confirming that the predictive content exists in the cross-section month by month.

Table 3. Panel predictive regressions of the short-variance payoff (H_1 and H_2).

	(1)	(2)	(3)
Dependent variable: payoff	H_1	H_2 horse race	Down + Up
VRP_total	0.204***	0.277***	—
	(0.036)	(0.064)	
VRP_down	—	−0.296**	−0.101
		(0.125)	(0.140)
VRP_up	—	—	0.087
			(0.100)
Intercept	−0.007	0.015	−0.004
	(0.006)	(0.006)	(0.012)
R^2	0.022	0.033	0.002
N	22,858	22,858	22,858

Notes. Pooled OLS estimates of equations (1) and (2) with month-clustered standard errors in parentheses. Column (1) is the H_1 specification; column (2) is the H_2 horse race; column (3) replaces the total premium with its downside and upside corridor legs. Once the total premium is included, the downside leg enters with a negative and significant coefficient (column 2); on their own, neither leg is significant (column 3). *, **, *** denote 10%, 5%, 1% significance.

Table 4. Fama-MacBeth cross-sectional regressions of the short-variance payoff.

Predictor	Mean coef.	t (NW)	p-value	Months
Model 1: payoff ~ Intercept + VRP_total				
Intercept	−0.004	−0.65	0.517	
VRP_total	0.141**	2.16	0.031	347
Model 2: payoff ~ Intercept + VRP_down				
Intercept	0.001	0.33	0.742	
VRP_down	−0.035	−0.46	0.648	347

Notes. Monthly cross-sectional OLS slopes averaged over time following Fama and MacBeth (1973), with Newey-West HAC t-statistics at three lags. The total-premium slope is significant at the 5% level, and the downside-premium slope is insignificant, in line with the panel evidence of Table 3. *, **, *** denote 10%, 5%, 1% significance.

Economically, a one-unit increase in annualized implied variance relative to trailing realized variance earns on average a short-variance payoff of 0.20 variance units higher over the following month. The null hypothesis of H_1 ($\beta = 0$) is therefore rejected, as the firm-level total VRP positively and significantly predicts the short-variance payoff. Section 5.6 shows,

however, that the pooled relationship is heavily influenced by the 2008-2009 crisis, while the Fama-MacBeth estimate is more stable.

5.2 The Downside Leg and the Total Premium (H_2)

H_2 predicts that the downside leg of the VRP drives the predictive content of the VRP, such that the VRP_total coefficient falls to zero once the downside leg is included. Column (2) of Table 3 shows the horse-race specification of equation (2). Against expectation, the VRP_total coefficient remains positive and highly significant (0.277, $p < 0.001$) when the downside leg is included, while the VRP_down slope turns negative and significant (−0.296, $p = 0.018$). For a given total premium, a firm with a larger downside premium therefore earns a lower rather than a higher payoff. The total premium is not subsumed by the downside leg at the firm level, the opposite of the index-level pattern documented in the literature. The null hypothesis of H_2 ($\beta_{dn} = 0$) is therefore not rejected, as the coefficient on VRP_down is negative rather than positive; the predictive content resides in the total premium, and not in its downside component.

Column (3) of Table 3 replaces the total premium with its upside and downside legs jointly. Neither coefficient is significant (−0.101 for VRP_down and 0.087 for VRP_up), and the R^2 falls to 0.002. Similarly, the Fama-MacBeth estimate on VRP_down in Table 4 is insignificant (−0.035, $t = -0.46$). Overall, the upside and downside components contain little predictive information individually at the firm level, and the total premium remains the stronger signal.

5.3 Stability Across Regimes (H_3)

H_3 predicts that the coefficient on the downside premium is constant across sample sub-periods. Table 5 reports equation (1) re-estimated with VRP_down. The coefficient is negative and insignificant in the pre-GFC sample (−0.094), close to zero in the GFC and post-GFC periods (0.002), and negative and insignificant in the modern period (−0.344). The test of equation (3) does not reject the null hypothesis of equal slopes across the three periods ($F = 0.28$, $p = 0.76$). The null hypothesis associated with H_3 is therefore not rejected, although none of the three sub-period coefficients are significant. Repeating the test with VRP_total does reject equality ($F = 4.58$, $p = 0.011$): the slope on the total premium is 0.250 ($t = 2.08$) in 1996-2007, 0.236 in 2008-2015, and -0.109 in 2016-2024, and the 2016-2024 slope is significantly lower than the 1996-2007 slope (difference = -0.359, $t = -2.19$). The predictive relationship for the total VRP therefore changes significantly across market regimes.

Table 5. Subsample stability of the downside-premium predictive slope (H_3).

Sub-period	β (VRP_down)	(SE)	t	N
Pre-GFC (1996–2007)	−0.094	(0.067)	−1.40	9,141
GFC / post-GFC (2008–2015)	0.002	(0.243)	0.01	6,441
Modern (2016–2024)	−0.344	(0.400)	−0.86	7,276

Notes. Estimated on three non-overlapping sub-periods, with month-clustered standard errors in parentheses. Wald test of equal slopes across the three sub-periods (pooled regression with sub-period interactions, month-clustered standard errors): F = 0.28, p = 0.76. *, **, *** denote 10%, 5%, 1% significance.

5.4 The Trading Strategy and its Risk-Adjusted Performance (H_4)

H_4 expects that a VRP-sorted calendar-time long-short portfolio generates positive risk-adjusted alpha after controlling for standard equity factor exposures, and aggregate variance risk. Table 6 reports the performance of the quintile portfolio sorted on VRP_total, with returns expressed as variance-swap returns so that all figures are in interpretable percentage units. The portfolio earns an annualized mean return of 0.786 (78.6%) against an annualized volatility of 1.234 (123.4%), corresponding to an annualized Sharpe ratio of 0.64. The return distribution is highly non-normal, with strong positive skewness (4.08) and excess kurtosis 41.3. The Sortino ratio is 1.31, while the 5% expected shortfall is −0.511 (−51.1%), and the maximum drawdown is −3.657 (−365.7%). These magnitudes reflect the levered character of a firm-level variance-selling strategy rather than realized dollar portfolio returns. The positive skewness is also consistent with the portfolio construction: although the strategy is short variance in the high-VRP quintile, it is also long variance in the bottom VRP quintile, whose returns are unbounded and can rise sharply during volatility spikes.

Table 6. Risk-adjusted performance of the long-short portfolios (H_4) and the IV-richness robustness.

	(1) VRP_total L-S	(2) VRP_rich L-S
Dependent variable: monthly long-short return		
Alpha	0.075***	0.086***
	(0.023)	(0.022)
MKT-RF	−1.127	0.781
	(0.878)	(0.792)
SMB	0.078	−0.697
	(0.726)	(0.917)
HML	−0.270	1.368
	(0.766)	(1.294)
RMW	0.441	−0.656
	(0.831)	(1.070)
CMA	−0.065	−0.889
	(1.152)	(1.338)
UMD	−0.598	−0.095
	(0.493)	(0.365)
ΔVIX	−0.011	0.008
	(0.008)	(0.012)
Annualised alpha	0.902	1.031
Annualised mean return	0.786	1.056
Annualised volatility	1.234	1.326
Annualised Sharpe ratio	0.637	0.796
Sortino ratio	1.313	1.093
Skewness	4.078	1.000
Excess kurtosis	41.299	30.172
Expected shortfall (5%)	−0.511	−0.675
Maximum drawdown	−3.657	−6.756
Observations (months)	347	336

Notes. Time-series Jensen-alpha regressions of the monthly equal-weighted quintile long-short variance-swap return on the Fama-French five factors, the Carhart momentum factor (UMD), and the change in the VIX (ΔVIX), estimated with Newey-West HAC standard errors at three lags (in parentheses). Column (1) sorts on the canonical premium VRP_total; column (2) sorts on the long-baseline signal VRP_rich. Returns and alphas are monthly variance-swap returns in decimal units. Both alphas

are significant at the 1% level and no factor loading is significant. The lower panel reports annualized performance and higher-moment risk diagnostics for both portfolios. *, **, *** denote 10%, 5%, 1% significance.

The Jensen alpha regression of equation (4) yields a significant annualized alpha of 0.902 (90.2%) with a Newey-West HAC t-statistic of 3.20. None of the seven factor loadings is statistically significant, indicating that the long-short returns are not explained by the equity and aggregate variance factors. The null hypothesis under H_4 is therefore rejected. The alpha also remains significant under both tighter 2.5/97.5 winsorization and no winsorization, and after transaction costs of up to twenty-five basis points per leg. These returns are expressed per unit of variance notional and do not impose a limit on losses at the individual-position level, so they should not be interpreted as directly investable portfolio returns. As a sensitivity test, realized variance is therefore capped at 2.5 times strike variance, which limits the loss of a single position to −150%. This cap binds in 3.25 percent of firm-months. Under this assumption, the strategy earns an annualized mean of 59.8% and a Sharpe ratio of 0.79, and an annualized alpha of 67.8% (t = 3.81), or 61.8% (t = 3.47) after costs of twenty-five basis points per leg. A looser cap of 6.25 times strike variance gives an alpha of 74.5% (t = 3.01). Measured in variance units rather than as a return, the long-short payoff earns an annualized alpha of 0.425 (t = 2.68). These cap levels are assumptions of this thesis rather than market conventions. Returns also remain concentrated, with the twelve best months accounting for 61.6 percent of the cumulative long-short performance.

5.5 OOS Predictive Ability

The in-sample results are complemented by the Welch-Goyal out-of-sample test in equation (5). Using training cutoffs at the end of 1999, 2002 and 2005, the out-of-sample R^2 is positive at 0.020, 0.020 and 0.020. Meaning the VRP-based forecast reduces the mean-squared prediction error by about two percent with respect to the historical mean-benchmark. However, the Clark-West test does not reject the null of no out-of-sample predictability at either the 5% or 10% level for any of the three cutoffs (one-sided p-values between 0.10 and 0.13). The out-of-sample evidence is therefore positive in sign but not statistically significant.

Table 7. Out-of-sample predictive ability across training cutoffs.

Training cutoff	R^2_OOS	Clark-West t	p (one-sided)
Panel A. Predictor = VRP_total			
Year-end 1999	0.020	1.27	0.102
Year-end 2002	0.020	1.16	0.123
Year-end 2005	0.020	1.15	0.126
Year-end 2015 (tested on 2016–2024)	−0.076	−0.77	0.780
Panel B. Predictor = VRP_rich (IV-richness)			
Year-end 2000	−0.005	—	—

Notes. Welch-Goyal out-of-sample R^2 of equation (5) relative to the prevailing-mean benchmark, with the one-sided Clark-West (2007) test of equal predictive accuracy. The Year-end 2015 row starts the expanding training window on 1996–2015 and evaluates the forecasts over 2016–2024 only; the negative R^2_OOS indicates that the model fails to predict payoffs in the modern era. The IV-richness alternative signal fails out-of-sample.

Setting the training cutoff at year-end 2015 yields a negative R^2 OOS of −0.076 (p = 0.780), indicating that the model does not beat the historical mean over 2016-2024. Combined with the significantly lower modern-era coefficient on VRP_total reported in Section 5.3, the evidence suggests that the VRP signal is concentrated in the earlier, crisis-heavy period rather than in the modern period, as the robustness checks also demonstrate. The IV-richness signal VRP_rich performs poorly out-of-sample with R^2 of −0.005 against the prevailing mean benchmark (Panel B of Table 7), which suggests that whatever predictive content there is lies in the implied-realized spread, and not in the long-baseline alternative.

5.6 Robustness Checks

The predictive slope on VRP_total in equation (1) is stable across the alternative specifications. Under two-way (firm × month) clustering, the coefficient remains 0.204 (t = 5.39), and with Driscoll and Kraay (1998) HAC standard errors at three lags the t-statistic is 4.43. A firm fixed-effects specification yields a coefficient of 0.194 (t = 5.53; within R^2 = 0.020), indicating the relationship holds within firms over time and is not an artefact of time-invariant differences in average VRP across firms. The result is also robust to macro controls. Adding the VIX level and its first difference to equation (1) gives a significant coefficient of 0.187, and additionally controlling for the term and default spreads gives 0.180 (t = 5.07). However, neither spread is statistically significant, highlighting that the predictive power remains after controlling for these market factors.

Table 8. Robustness of the H_1 predictive slope across inference and specification choices.

Specification	Coef.	t	R^2 / note
Baseline (one-way cluster on month)	0.204	5.60	$R^2 = 0.022$
Two-way cluster (firm × month)	0.204	5.39	$p < 0.001$
Driscoll–Kraay HAC (3 lags)	0.204	4.43	$p < 0.001$
Firm fixed effects	0.194	5.53	within $R^2 = 0.020$
+ Macro-state controls (VIX, ΔVIX)	0.187	5.04	$R^2 = 0.032$
+ VIX, ΔVIX, term and default spread	0.180	5.07	$R^2 = 0.034$

Notes. Coefficient on VRP_total in equation (1) under alternative inference and specification choices, with month-clustered standard errors unless stated otherwise. Two-way clustering follows Cameron et al. (2011). The term spread is the 10-year minus 3-year Treasury yield and the default spread the Moody's Baa minus Aaa corporate bond yield (frb.rates_daily). All specifications use the 22,858 firm-months of the main panel.

Table 9 applies a transaction cost overlay of ten and twenty-five basis points, per leg, to the long-short return before re-estimating alpha of equation (4). The monthly alpha lowers from 0.075 at no cost, to 0.073 at 10 basis points and 0.070 at twenty-five basis points per leg (equivalent annualized alphas of 90.2%, 87.8% and 84.2%). Alpha stays statistically significant at the 1% level throughout, showing that the abnormal performance of the strategy remains robust to these assumed costs.

Table 9. Transaction-cost overlay on the long-short Jensen alpha.

Per-leg cost	Monthly α	t	Annualised α
0 basis points	0.075***	3.20	0.902
10 basis points	0.073***	3.11	0.878
25 basis points	0.070***	2.99	0.842

Notes. Monthly Jensen alpha of the VRP_total long-short portfolio, net of a transaction cost charged on each leg every month, estimated by equation (4) with Newey–West HAC standard errors at three lags. *** denotes significance at the 1% level.

Replacing the premium with the long-baseline richness signal VRP_rich = IV_var_total – HV_long252d, following the approach of Goyal & Saretto (2009), produces a weaker predictor in the panel regressions: on its own, VRP_rich enters with a negative coefficient (−0.238, t = −2.16). In a horse race VRP_total stays positive and significant with a coefficient of 0.196 (t = 5.16) while VRP_rich stays negative (−0.213, t = −2.21). Its out-of-sample R^2 is also negative. The calendar-time long-short strategy sorted on VRP_rich nonetheless earns a

higher alpha than the canonical strategy (103.1% annualized, t = 3.86; Table 6). The portfolio and panel-regression results for VRP_rich therefore conflict. The two tests are not directly comparable, however, as the panel regression uses the payoff in variance units, while the portfolio ranks firms and measures performance as a variance-swap return, which scales each payoff by its implied variance. The implied-realized spread is therefore the more reliable predictor on the panel-regression and OOS metrics, while the long-baseline signal performs better only on the portfolio metric.

Table 10. Alternative long-baseline signal: panel regressions on VRP_rich.

	(1) Standalone	**(2) Horse race**
Dependent variable: payoff		
VRP_total	—	0.196***
		(0.038)
VRP_rich	−0.238**	−0.213**
	(0.110)	(0.096)
Intercept	−0.010	−0.009
	(0.007)	(0.007)
R^2	0.010	0.030
Observations	22,061	22,061

Notes. Pooled OLS with month-clustered standard errors in parentheses. Column (1) is the standalone panel regression on VRP_rich = IV_var_total − HV_long252d (Goyal & Saretto, 2009); column (2) is the horse race against the canonical VRP_total. Payoff, VRP_total and VRP_rich are winsorized within month at the 1st and 99th percentile, as in the main analysis. *, **, *** denote significance at the 10%, 5% and 1% level.

Two additional structural-break tests refine the sub-sample analysis of Section 5.3. Decomposing the crisis-heavy sample into an acute phase (2008-2009), a recovery phase (2010-2012) and a calm phase (2013-2015), with an additional row for the COVID-19 episode in 2020 (estimated on twelve months), the predictive slope on VRP_total is highly significant during the acute phase ($\beta = 0.217$, t = 8.79), though with an R^2 of only 0.025, marginally significant in the recovery phase ($\beta = 0.166$, t = 1.72), and insignificant in the calm and COVID periods. Splitting the panel on the VIX state, where the high-VIX months are the top tercile of the VIX distribution, the slope is 0.196 (t = 5.08) in high-VIX months and 0.255 (t = 1.46) in low/mid-VIX months. The interaction term VRP_total × HighVIX is insignificant (−0.059, t = −0.33, p

= 0.74), so the null hypothesis of equal slopes across volatility states ($\beta_2 = 0$) cannot be rejected. However, the relationship is estimated much more precisely during high-VIX months, where the R^2 is 0.054 compared with only 0.003 in the two bottom terciles. The results therefore do not show that the slope differs across volatility states, but they do not establish that the relationship is equally strong across them. The same decomposition for VRP_down (Panel D of Table 11) shows a different pattern. Its coefficient shows no significant slope in the acute phase (0.080, t = 0.35), but positive and significant slopes in the recovery (0.277, t = 3.05) and calm (0.135, t = 2.17) phases. When 2008-2015 is estimated as one period, the slope on VRP_down is near zero (Table 5), so these sub-periods reveal positive post-crisis relationships that are hidden when the years are pooled together.

Meanwhile, the estimate for 2020 is insignificant and based only on twelve months, which is consistent with the idea that a volatile market alone does not guarantee a positive payoff: short-variance positions win when elevated volatility mean-reverts, as was the case in 2008-2009, and lose when it persists.

Table 11. Structural-break analysis of the predictive slopes on VRP_total and VRP_down.

	β	(SE)	t	R^2
Panel A. Crisis-era decomposition				
GFC acute (2008–2009)	0.217***	(0.025)	8.79	0.025
Post-GFC recovery (2010–2012)	0.166*	(0.096)	1.72	0.025
Calm / QE (2013–2015)	0.031	(0.037)	0.84	0.001
COVID (2020)	−0.228	(0.207)	−1.10	0.037
Panel B. VIX-state split (VIX at portfolio formation, full-sample tercile)				
High-VIX (top tercile)	0.196***	(0.039)	5.08	0.054
Low / mid-VIX	0.255	(0.174)	1.46	0.003
Panel C. VIX interaction test (R^2 = 0.024, N = 22,858)				
VRP_total	0.255	(0.174)	1.46	
VRP_total × HighVIX	−0.059	(0.178)	−0.33	
Panel D. Crisis-era decomposition, slope on VRP_down				
GFC acute (2008–2009)	0.080	(0.229)	0.35	0.001
Post-GFC recovery (2010–2012)	0.277***	(0.091)	3.05	0.049
Calm / QE (2013–2015)	0.135**	(0.062)	2.17	0.011
COVID (2020)	−0.703	(0.692)	−1.02	0.078

Notes. Panel A re-estimates the H_1 panel regression on calendar sub-windows; Panel B splits on the VIX known at portfolio formation, using the top tercile of the full-sample VIX distribution as threshold; Panel C is the interaction specification of equation (6); Panel D repeats Panel A with VRP_down as the predictor. Standard errors clustered by month in parentheses. Observations in Panels A and D: 1,605, 2,389, 2,447 and 816 respectively; in Panel B: 7,633 and 15,225. *, **, *** denote significance at the 10%, 5% and 1% level.

The long-short alpha is further re-estimated under no winsorization and a tighter 2.5/97.5% trimming, next to the 1/99% baseline. The annualized alpha ranges from 0.861 under the tightest trimming to 0.902 in the baseline, with 0.894 in the untrimmed specification. Alpha remains significant in all three treatments.

Table 12. Winsorisation robustness of the long-short alpha.

Treatment	Ann. ret.	Sharpe	Monthly α	Ann. α
None (classic)	0.764	0.560	0.075***	0.894
1/99% (baseline)	0.786	0.637	0.075***	0.902
2.5/97.5% (tight)	0.758	0.783	0.072***	0.861

Notes. VRP_total long-short under three within-month winsorisation thresholds. Monthly-alpha t-statistics are 2.74, 3.20 and 3.77 respectively. *** denotes significance at the 1% level.

Because the structural-break results point to the 2008-2009 crisis, the main specifications are re-estimated without this period (Table 13). Excluding 2008-2009 lowers the pooled coefficient of VRP_total from 0.204 to 0.094, which is no longer statistically significant ($t = 1.06$), with an R^2 of 0.005; excluding only the crash months from September 2008 to March 2009 gives a similar result (0.111, $t = 1.27$). A small number of months carry a large part of the full-sample estimate: September 2008 alone accounts for 47.6 percent of the covariance that underlies the pooled slope, and the five most influential months, all in 2008 (with the exception of November 2001), account for 89.1 percent. In September 2008, the cross-sectional standard deviation of VRP_total was 2.57, 53.6 times its level in the median month (0.048). However, the result is less sensitive when each month is equally weighted. The Fama-MacBeth coefficient stays positive and significant after excluding the 2008-2009 period (0.082, $t = 2.88$). The long-short strategy also holds up: its annualized mean return is 66.0% with a Sharpe ratio of 0.76, while annualized alpha falls from 90.2% to 70.7% ($t = 3.28$). Removing only the September 2008-March 2009 crash period gives a similar alpha of 72.5% ($t = 2.96$). In the horse race specification, VRP_total is no longer significant at the 5% level after excluding 2008-2009

(0.154, t = 1.90), whereas the negative coefficient on VRP_down stays significant (−0.204, t = −2.30).

Table 13. Main results with and without the 2008–2009 crisis.

	Full sample	Excl. 2008–2009	Excl. Sep 2008–Mar 2009
Pooled slope on VRP_total (H_1)	0.204*** (5.60)	0.094 (1.06)	0.111 (1.27)
R^2 (H_1)	0.022	0.005	0.002
Fama-MacBeth slope on VRP_total	0.141** (2.16)	0.082*** (2.88)	—
Horse race: VRP_total	0.277*** (4.33)	0.154* (1.90)	0.183* (1.88)
Horse race: VRP_down	−0.296** (−2.37)	−0.204** (−2.30)	−0.304* (−1.84)
Long-short annualized return	0.786	0.660	0.687
Long-short Sharpe ratio	0.637	0.757	0.587
Annualized alpha	0.902*** (3.20)	0.707*** (3.28)	0.725*** (2.96)
Firm-months	22,858	21,253	22,392

Notes. t-statistics in parentheses: month-clustered for the pooled and horse-race regressions, Newey-West (three lags) for the Fama-MacBeth slope and the alpha. The long-short return and alpha are annualized variance-swap returns. The Fama-MacBeth slope is not re-estimated for the crash-month exclusion. *, **, *** denote significance at the 10%, 5% and 1% level.

Two further checks concern the sample and the moneyness of the signal. Removing the ten distressed firms described in Section 3.1 lowers the pooled VRP_total slope to 0.111 (t = 1.81) and the annualized alpha to 65.9% (t = 2.63), so these firms strengthen the results, most clearly the pooled slope, which without them is significant only at the 10% level. Finally, a tail-only version of the premium, in which implied variance is integrated over the $|\Delta| \in [10, 25]$ wings of the surface only while the payoff is left unchanged, gives a pooled slope of 0.197 (t = 5.61), a Fama-MacBeth slope of 0.112 (t = 2.49) and a long-short alpha of 89.8% (t = 3.23). This check has limited power, as the tail-only premium has a correlation of 0.994 with VRP_total, but it indicates that the results do not depend on the near-the-money segment of the surface.

6 Discussion

The first and central finding of this study that the firm-level VRP positively and significantly predicts the subsequent short-variance payoff, is consistent with the literature that the VRP contains return-predictive information. Bollerslev et al. (2009) established this at the

index level, and this result extends this predictability to the firm-level cross-section to a variance-swap style payoff rather than to individual stock or index returns. The results complement the firm-level evidence found by Bakshi & Kapadia (2003), An et al. (2014), Bali & Zhou (2016). However, the explanatory power is modest. The coefficient of 0.204 comes with an in-sample R2 of 0.022, well below the quarterly R2 of 7 to 8 percent that Bollerslev et al. (2009) report for the aggregate market, although the horizon and the unit of analysis differ. The pooled slope is also no longer significant once 2008-2009 is excluded, whereas the equally weighted Fama-MacBeth slope remains significant. The predictability is therefore best described as a small cross-sectional effect that survives equal weighting of months, with a pooled magnitude that is dominated by the most turbulent months.

The second finding, however, diverges sharply from the literature. Feunou et al. (2018) show that, at the index level, the downside VRP is the engine of predictability, and the upside component is washed out in a joint specification. This study finds the opposite at the firm-level: the total premium is not absorbed by the downside leg, and once the total premium is controlled for, a higher downside premium predicts a lower rather than a higher payoff, also when 2008-2009 is excluded. On their own, neither the downside nor the upside leg predicts the payoff. A plausible interpretation that is consistent with the tail risk evidence of Andersen et al. (2020) and Todorov (2010), is that the downside premium at the index level is dominated by the price of systematic market-crash insurance, which has no clear counterpart at the firm level. At the firm level, a high downside premium relative to the total premium may instead partly reflect a rational upward adjustment in the probability of firm-specific downside events, as set out in Section 2.3.1, which the short-variance position then pays for when they materialise. The acute phase of the financial crisis is consistent with this interpretation: the ten distressed firms, whose own distress was being priced, are responsible for 25.3 percent of the worst one percent of firm-month payoffs while making up only 10.1 percent of all observations. This finding highlights the limits to the generality of the Feunou et al. (2018) interpretation rather than contradicting its index-level mechanism.

Thirdly, the downside leg has no significant predictive slope in any of the three sub-periods, and the null hypothesis of equal slopes cannot be rejected. The shorter windows of Table 11 investigate this further: within 2008-2015, the coefficient on VRP_down is positive and significant in both the recovery (0.277, t = 3.05) and calm (0.135, t = 2.17) phases. The total premium is less stable: its slope is significantly lower in 2016-2024 than in 1996-2007, and the structural-break analysis shows that the pooled evidence is concentrated in the acute phase of the financial crisis, when the cross-sectional dispersion of the VRP was more than fifty

times its median level. Conditioning on the VIX state gives similar point estimates in high-VIX (0.196) and low-VIX (0.255) months, and the interaction test does not reject the null hypothesis of equal slopes (−0.059, t = −0.33), but the slope is only precisely estimated in high-VIX months, where the R^2 is 0.054 against 0.003. The negative, though insignificant, COVID point estimate is consistent with the view that a high volatility level alone does not guarantee a positive payoff: short-variance positions win when elevated volatility mean-reverts, as in 2008-2009, and lose when it persists, as in the acute COVID window. One interpretation is that the VRP partly captures time-varying compensation for uncertainty and adverse tail risk: when investors become more concerned about negative outcomes, implied variance can increase relative to subsequently realized variance as the demand for volatility protection increases. This resonates with the regime-dependent characteristic of the VRP predictability found by Pyun (2019) and Andersen et al. (2020).

The last finding of the long-short strategy earning significant abnormal returns that survive standard factor controls and transaction costs, aligns with the evidence that variance-selling strategies have historically earned positive risk-adjusted returns that conventional equity factors cannot explain. This aligns with Bondarenko (2014), who found this at the index level, and DeMiguel et al. (2013). The substantial annualized alpha of ninety percent reflects the highly levered character of a variance-selling strategy expressed as a variance-swap return at the firm-level; but the Sharpe ratio of 0.64, is the more conservative summary of risk-adjusted performance, and the heavy tails (a maximum drawdown of more than three and a half times capital) are the economic price paid for it. Moreover, the alpha is not driven by the crisis alone, as it remains at 70.7% without 2008-2009, and at 67.8% when the loss on a single position is capped under the assumption described in Section 5.4. That the alpha is not captured by a change in the VIX, suggests that the firm-level signal captures cross-sectional variation in variance pricing, consistent with Cao & Han (2013).

Taken together, the results support the market-friction interpretation of the VRP over a purely systematic-risk interpretation, where the frictions are those of the option market, since frictions in the underlying equity market cannot be separately identified with this design: the signal predicts realized payoffs in the cross-section and generates alpha that is not removed by factor exposures and transaction costs. The dominance of the total over the downside premium and the survival of the alpha net of transaction costs are also difficult to associate with the premium being fair compensation for systematic crash risk alone. At the same time, the insignificant out-of-sample evidence, the modest explanatory power and the heavy tails of the

long-short portfolio returns are a reminder that the strategy's profitability is susceptible to rare but severe variance spikes.

7 Conclusion

This thesis examines whether the variance risk premium predicts the risk-adjusted returns of short-variance strategies in U.S. large-cap equities. The variance risk premium has become a central object in asset pricing as it reflects the compensation investors demand for bearing volatility risk. Nonetheless, most existing evidence treats it as a macroeconomic predictor of aggregate returns rather than as a firm-level signal for a direction-agnostic volatility strategy. This study addresses this gap by constructing the premium firm by firm and investigating whether it predicts realized payoffs for a short-variance position in a firm. This leads to the central research question: does the firm-level variance risk premium predict the risk-adjusted returns of short-variance strategies in U.S. large-cap equities?

To answer this question, the study uses an unbalanced firm-month panel of seventy-three large capitalization U.S. equities between 1996 and 2024, including ten firms that failed or became severely distressed. Option-implied variance is constructed from the OptionMetrics standardized volatility surface, and the realized variance is calculated from CRSP daily returns. The total VRP and its downside and upside components are estimated for each firm-month. The four hypotheses are tested using pooled panel predictive regressions with month-clustered standard errors, complementary Fama-MacBeth regressions, calendar-time quintile long-short portfolio sorts evaluated by Jensen alpha, and a Welch-Goyal out-of-sample forecasting exercise. In-sample, the total VRP positively and significantly predicted the short-variance payoff, with a panel coefficient of 0.204 ($t = 5.60$), but with an R^2 of only 2.2%. The total VRP is strongly influenced by the 2008-2009 crisis, while the Fama-MacBeth slope remains significant after excluding it from the sample. Contrary to the index-level literature, the downside leg does not subsume the total premium. Conditional on the total VRP, a higher downside premium predicts lower rather than higher payoffs and its slope is insignificant in each of the three H_3 sub-periods, although positive and significant in the shorter 2010-2012 and 2013-2015 windows. In terms of the formal tests, the null hypothesis is rejected for H_1 ($\beta = 0$) and for H_4 ($\alpha = 0$), and not rejected for H_2 ($\beta_{dn} = 0$) and for H_3 (equal downside slopes across the three sub-periods).

The slope on the total VRP does not differ significantly between high and low/mid volatility months, but it is only significant in high-volatility months. The long-short strategy generates an annualised Jensen alpha of approximately 90% (a monthly alpha of 7.51%, t = 3.20) and an annualised Sharpe ratio of 0.64 that remain significant after standard factor controls, plausible transaction costs, alternative outlier treatments and the exclusion of 2008-2009. Out of sample, the VRP improves on the prevailing mean by about two percent of mean-squared prediction error, but this improvement is not statistically significant. Forecast performance also turns negative when the forecasts are restricted to 2016-2024.

Three key takeaways stand out from this study of the VRP. First, the predictability documented at the index level extends to the firm-level and to variance-swap style payoffs, although with modest explanatory power. Its internal structure, however, does not carry over in the same way: the downside dominance found at the index level is not observed for individual firms, where the total premium is the more dominant signal. Second, trading on the firm-level VRP by selling variance in the highest-VRP firms and buying in the lowest-VRP firms, generates significant abnormal returns. The resulting alpha is not explained by standard equity and aggregate volatility factors and remains significant after transaction costs, suggesting that the returns cannot solely be attributed to systematic risk exposures. Third, the signal is regime dependent: its statistical strength is concentrated in periods of elevated volatility, so its practical value is closely linked to the rare but severe drawdowns faced by variance sellers that they must be willing to absorb.

7.1 Limitations and Future Research

Several limitations of the study affect the reliability of the conclusions and point to some productive extensions. First, the trailing realized variance computed from daily returns is used as the conditional-variance proxy, while implied variance is taken from an interpolated volatility surface on a fixed delta grid rather than directly from traded contracts. Measurement error thus enters both sides of the premium; replacing the daily proxy with a high-frequency realized-variance measure, as used by Feunou et al. (2018), could provide a more precise measure and enable the predictive results to be tested under a different specification. Second, the cross-section is limited to seventy-three firms and was selected by hand. Consisting of prominent U.S. firms, several of which are among today's largest companies, with ten distressed firms added. The sample is therefore not representative of the wider universe of optionable equities and remain susceptible to selection and survivorship bias. The relatively small cross-section also means that the quintile portfolios contain few firms. The scope of the sample and

the use of daily rather than intraday data were also constrained by available computing power. Nonetheless, extending the analysis to the full universe of optionable U.S. equities would strengthen the analysis. Third, the alpha is expressed as a variance-swap return, which produces large and heavy-tailed percentage returns at the firm level, as implied variance can be small for low-volatility firms. The cap levels used to translate the variance-swap returns into a more investable measure are assumptions rather than observed contract terms. A fuller treatment of the realized return net of bid-ask spreads on the underlying option contracts, in the spirit of Bondarenko (2014), would strengthen their economic interpretability. The downside and upside legs are constructed as averages over one wing of the surface, meaning they do not add up to the total premium. A corridor construction in which they do, as in Feunou et al. (2018), would enable a more direct firm-level test of the downside decomposition. Lastly, a structural-break analysis examining the post-2016 growth in single-name option volumes as a possible reason for weaker predictability in the modern period could provide further insight into the state dependence of the firm-level VRP.

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Declaration on the use of generative AI

I used generative AI tools, including Claude (Anthropic), ChatGPT (OpenAI), and Gemini (Google), during the preparation of this thesis. Their use fell into four categories. First, for coding and debugging, they were used extensively to help develop, run, debug and adjust the Python code necessary for the empirical analysis on WRDS data. Second, I used them to inspect the underlying data and verify that the correct data were being used and that the inputs and outputs were consistent with the code and intended analysis. Third, they assisted with the formatting of tables, the appendix and mathematical formulas. Fourth, I used them to summarize relevant literature, compare and evaluate the methodological approaches used in prior studies, and identify the approaches most appropriate for this study. I also used them to brainstorm ways to assess, adapt and improve these methodologies before independently deciding which methods to retain and apply.

Appendices

Table 14. Appendix A: raw data inventory — source databases and fields retrieved from WRDS.

Field	Type	Description and use in the empirical analysis
Panel A. CRSP stock names and monthly stock file — identifier resolution [crsp.stocknames, crsp.msf]		
ticker	string	Ticker symbol, with the dates (namedt, nameenddt) on which each CRSP security carried it; used to find every security that carried a list ticker between 1996 and 2024.
permno, shrcd	integer	CRSP permanent number and share code; only ordinary shares (share codes 10, 11 and 12) are eligible, and where several securities carried a ticker the one with the largest average market equity (crsp.msf) while carrying it is retained (Section 3.1).
Panel B. OptionMetrics IvyDB U.S. — standardised volatility surface [*optionm_all.vsurfd{YYYY} (yearly tables, 1996–2024)*]		
secid	integer	OptionMetrics security identifier; linked to the CRSP permno via Panel C.
date	date	Trading date of the surface observation.
days	integer	Time to maturity in calendar days; restricted to days = 30 throughout the analysis to fix the implied-variance horizon.
delta	numeric	Standardised option delta on the OptionMetrics surface grid; signed (positive for calls, negative for puts), with absolute deltas in {10, 15, 20, … , 80, 85, 90} on the 30-day slice.
cp_flag	string	Call/put indicator; “C” for the call wing, “P” for the put wing of the surface.
impl_volatility	numeric	Black-Scholes implied volatility quoted on the standardised delta grid; squared and integrated over delta to construct iv_var_total, iv_var_down and iv_var_up (Section 3.2.1).

Panel C. WRDS OptionMetrics–CRSP linking table [*wrdsapps.opcrsphist*]		
secid, sdate, edate	integer, date	OptionMetrics security identifier and the dates for which its link to a CRSP permno is valid; surface observations outside the link window are dropped.
permno	integer	CRSP permanent number; primary key into crsp.dsf, used to merge option-implied measures with stock-return data.
Panel D. CRSP daily stock file [*crsp.dsf*]		
permno	integer	CRSP permanent number; merge key against Panel C.
date	date	Trading date of the price observation.
ret	numeric	Daily holding-period return; squared and summed over the trailing (forward) 21-trading-day window to construct rv_back_21d (rv_fwd) and over signed sub-windows for rs_back_down / _up and rs_fwd_down / _up (Section 3.2.2).
prc	numeric	Closing price (negative when bid-ask midpoint is reported); used in the market-equity construction me = |prc| × shrout.
shrout	numeric	Shares outstanding (in thousands); combined with prc to yield market equity me used as a firm-size descriptive.
vol	numeric	Daily share-trading volume; retained as a liquidity descriptive.
Panel E. Fama–French research factor files [*ff_all.fivefactors_daily, ff_all.factors_daily*]		
date	date	Trading date of the factor observation.
mktrf	numeric	Excess market return; included in the long-short alpha regression in equation (4).
smb	numeric	Fama–French (2015) size factor.
hml	numeric	Fama–French (2015) value factor.
rmw	numeric	Fama–French (2015) profitability factor.
cma	numeric	Fama–French (2015) investment factor.
umd	numeric	Carhart (1997) momentum factor (Up-Minus-Down); pulled from ff_all.factors_daily and joined to the FF5 file by date.
rf	numeric	Risk-free rate (one-month Treasury); used to construct excess returns where required.
Panel F. CBOE volatility index file [*cboe_all.cboe*]		
date	date	Trading date of the VIX observation.
vix	numeric	CBOE S&P 500 implied-volatility index (close), in volatility points.
dvix	numeric (constructed)	First difference of vix, computed as vix.diff(); enters the alpha regression in equation (4) as a parsimonious proxy for innovations in aggregate variance risk.
Panel G. Federal Reserve interest rates [frb.rates_daily]		
dgs10, dgs3, daaa, dbaa	numeric	10-year and 3-year Treasury yields and Moody's Aaa and Baa corporate bond yields; term spread = dgs10 − dgs3 and default spread = dbaa − daaa (Section 3.3).

Notes. The table inventories every raw field retrieved from WRDS by the script and the use of each field in the construction of the analysis variables defined in Sections 3.2.1 to 3.3. Field names are reproduced in italics as they appear in the underlying WRDS tables. The OptionMetrics IvyDB volatility surface is queried year by year through the yearly vsurfd{YYYY} tables, restricted to days = 30 to fix the implied-variance horizon to one calendar month. The CRSP daily file is queried over the full sample. Each ticker is resolved to one CRSP permno before any option or return data are merged (Panel A), and all raw pulls are stored so that the analysis can be reproduced exactly. Fama & French (2015) five factors are joined to the Carhart (1997) momentum factor by trading date, and the CBOE VIX index is differenced to obtain dvix. All time-series are aligned on calendar dates and aggregated to the firm-month frequency before merging.